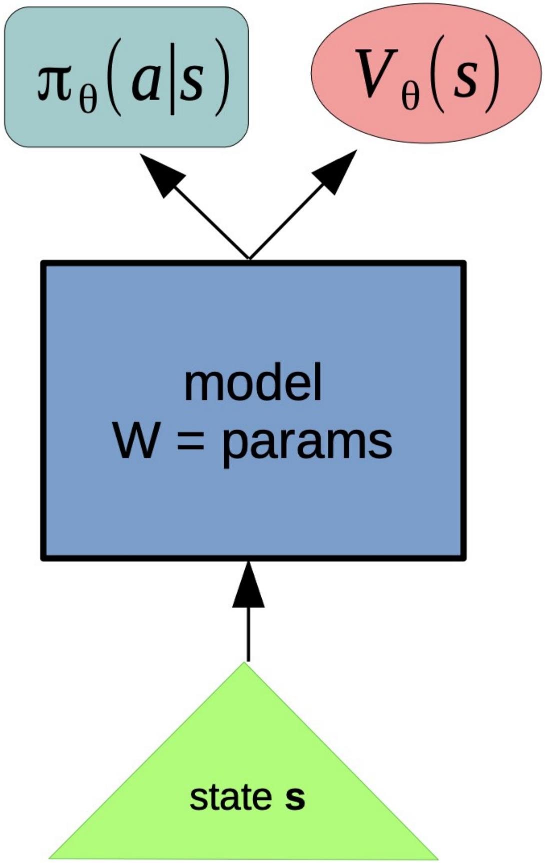
A2C homework tips



week06_policy_based/a2c-optional.ipynb

- gym 0.17 or 0.18
- no Windows

YSDA/HSE anq

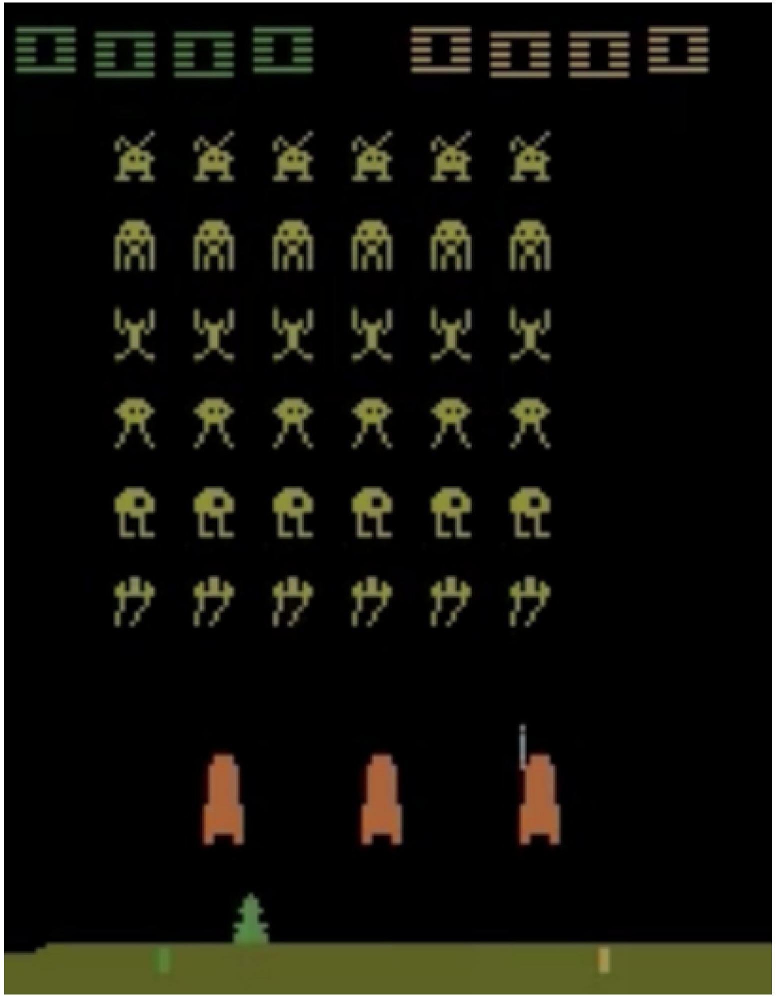
Install atari on gym 0.17

If pip3 install gym[accept-rom-license] doesn’t work:

- pip install gym[atari]==0.17.*
- Download ROMs from http://www.atarimania.com/roms/Roms.rar
- Extract it, there will be two folders: “HC ROMS” and “ROMS”
- python -m atari_py.import_roms ROMS
- If everything is ok, there will be a lot of “copying ...” in the output

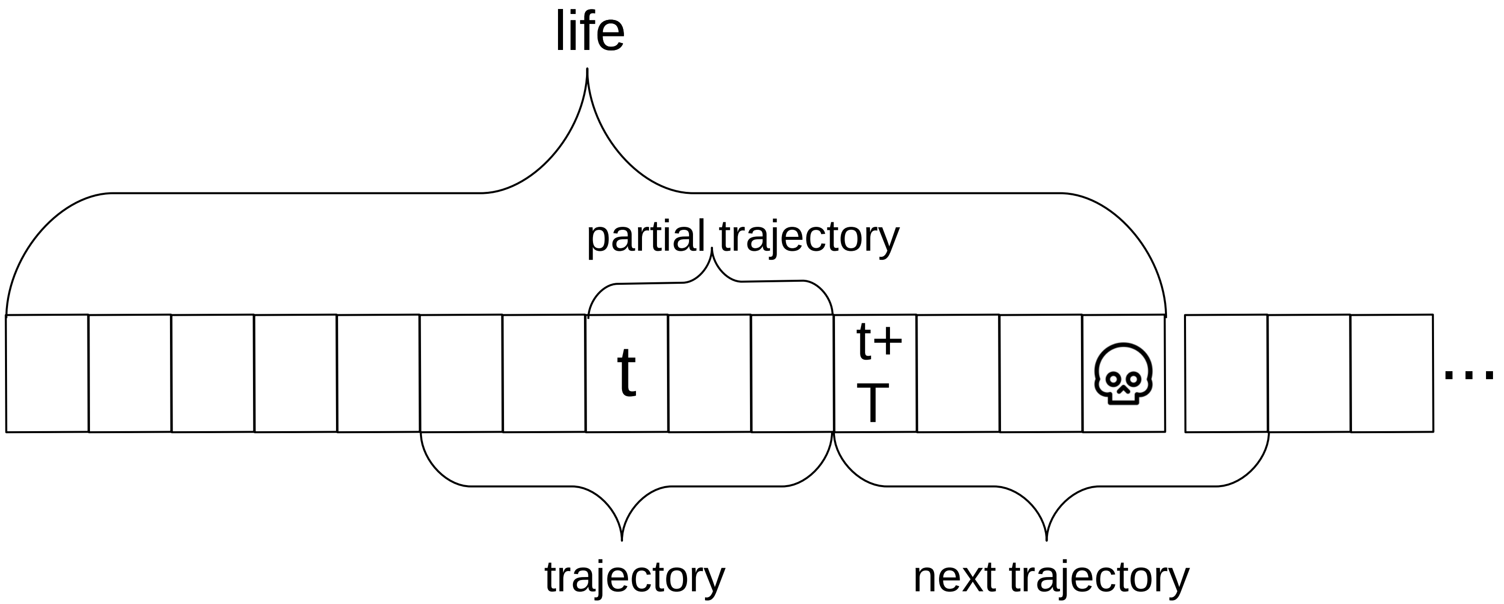
You will implement

- Network architecture (NatureDQN with 2 heads)
- Policy (wrapper for model)
- ComputeValueTargets
- MergeTimeBatch
- Loss (policy, value, entropy)
- Training loop and plots



SpaceInvadersNoFrameskip-v4

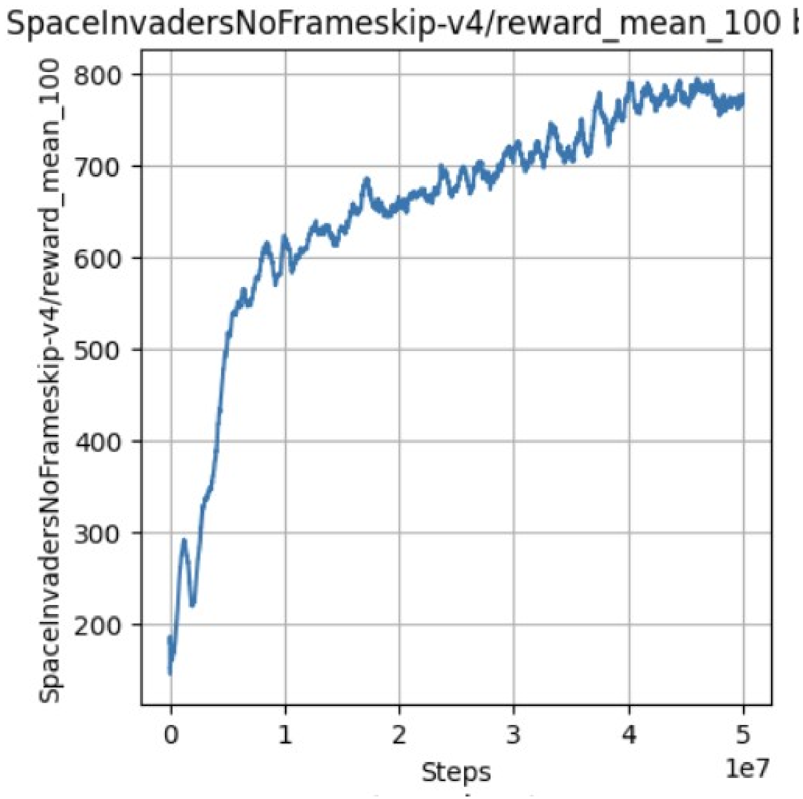
Definitions in ComputeValueTargets



len(trajectory) is nsteps
len(partial trajectory) is T

What is our target score

- Average raw reward over last 100 episodes
- Raw, not clipped!
- Average over 100, not just last one!
- Don’t compute manually!
- NumpySummaries.get_values(“SpaceInvadersNoFrameskip-v4/reward_mean_100”)
- Target score (to pass the task) is 600
- To get non-zero mark: at least 450



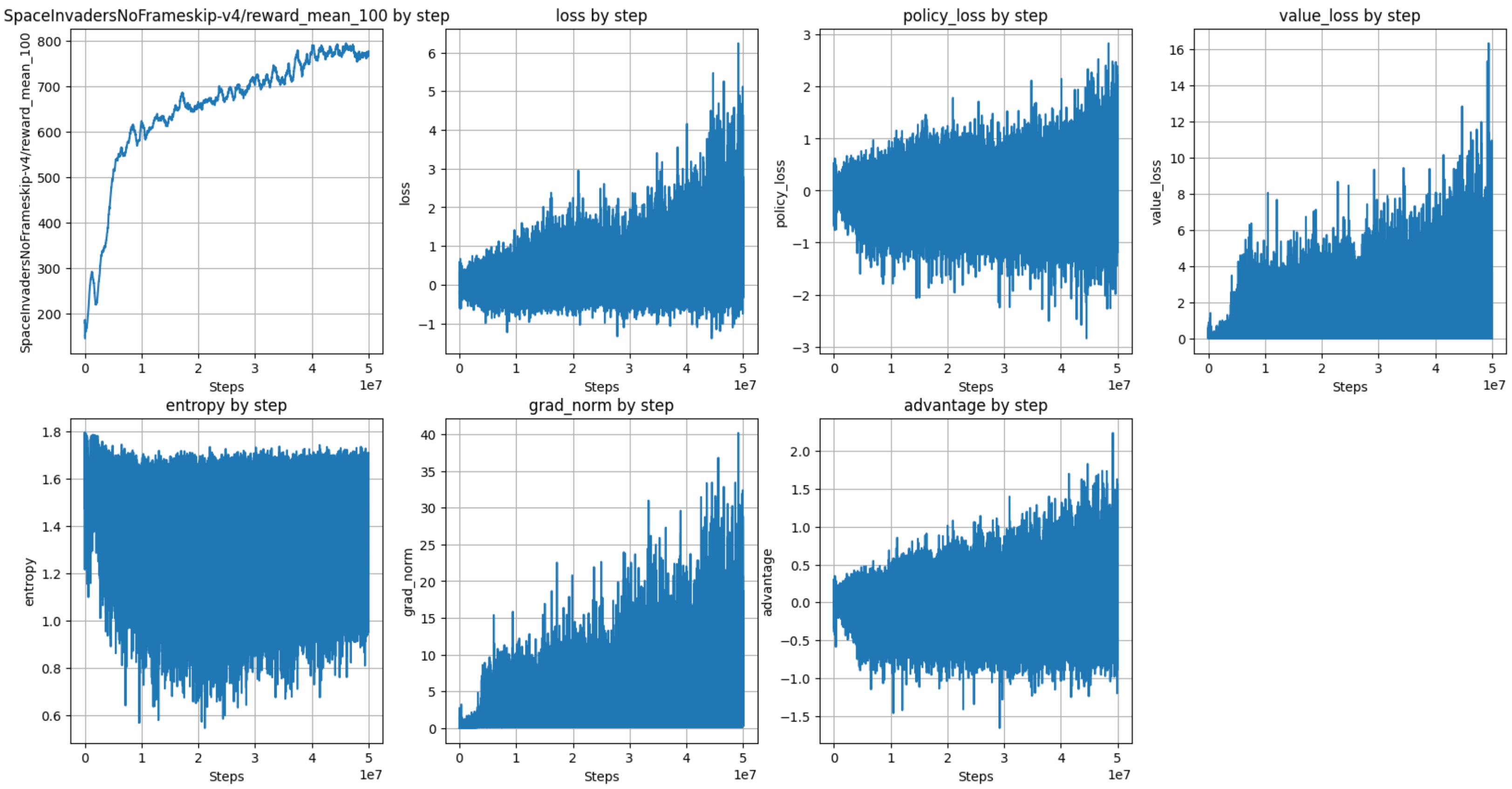
How to use NumpySummaries

- It’s the code that computes raw rewards for your plots
- env = nature_dqn_env("SpaceInvadersNoFrameskip-v4", nenvs=8, summaries='Numpy')
- Before calling runner.get_next() call NumpySummaries.set_step(runner.step_var)
- And then, after calling runner.get_next() you can get values by calling NumpySummaries.get_values(“SpaceInvadersNoFrameskip-v4/reward_mean_100”)

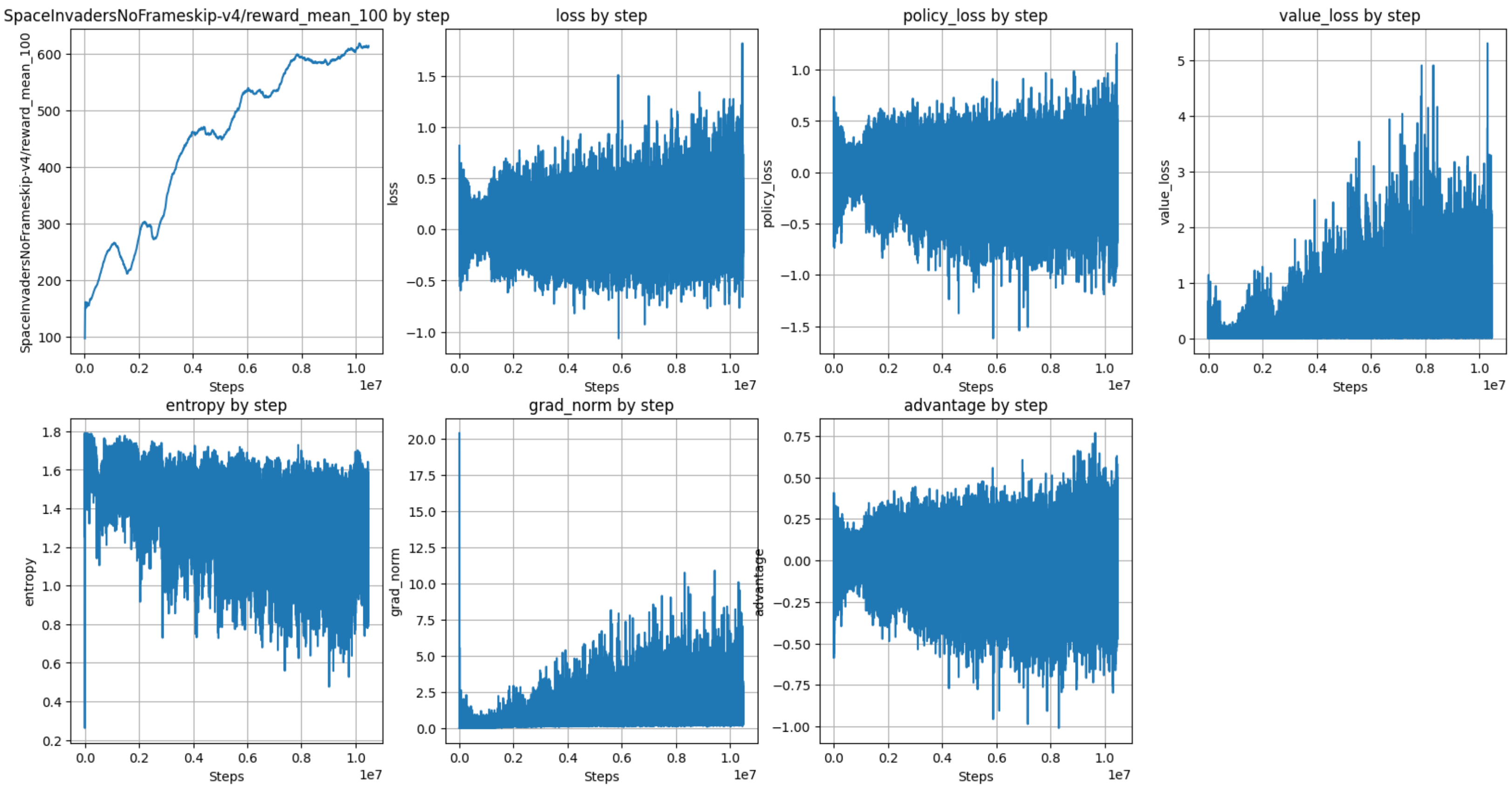
Custom values in NumpySummaries

- smth_eg_policy_loss = …
- env.add_summary_scalar('policy_loss', smth_eg_policy_loss.item())
- After calling runner.get_next() you can get values by calling NumpySummaries.get_values(“policy_loss”)

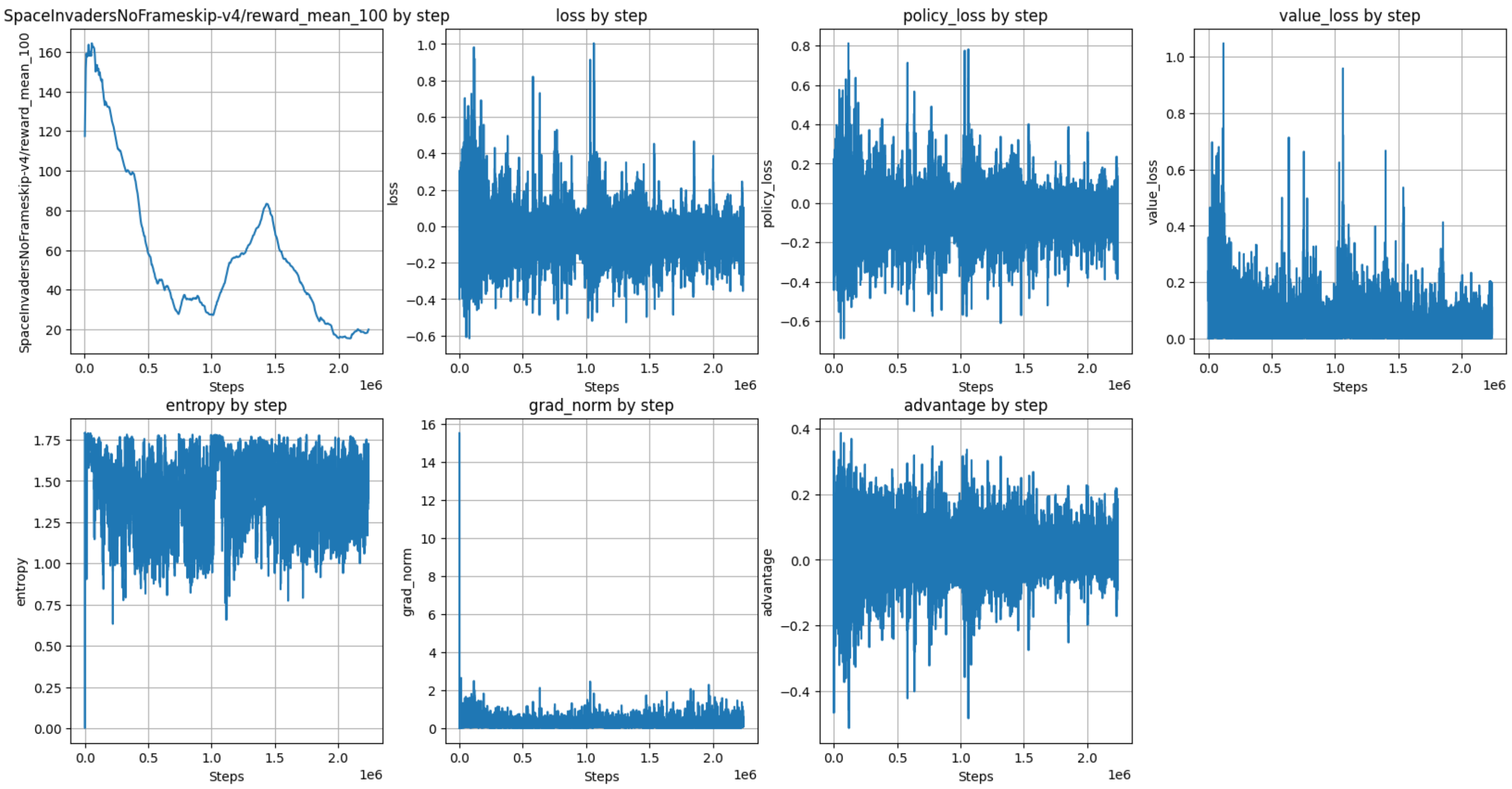
Example plots: correct solution



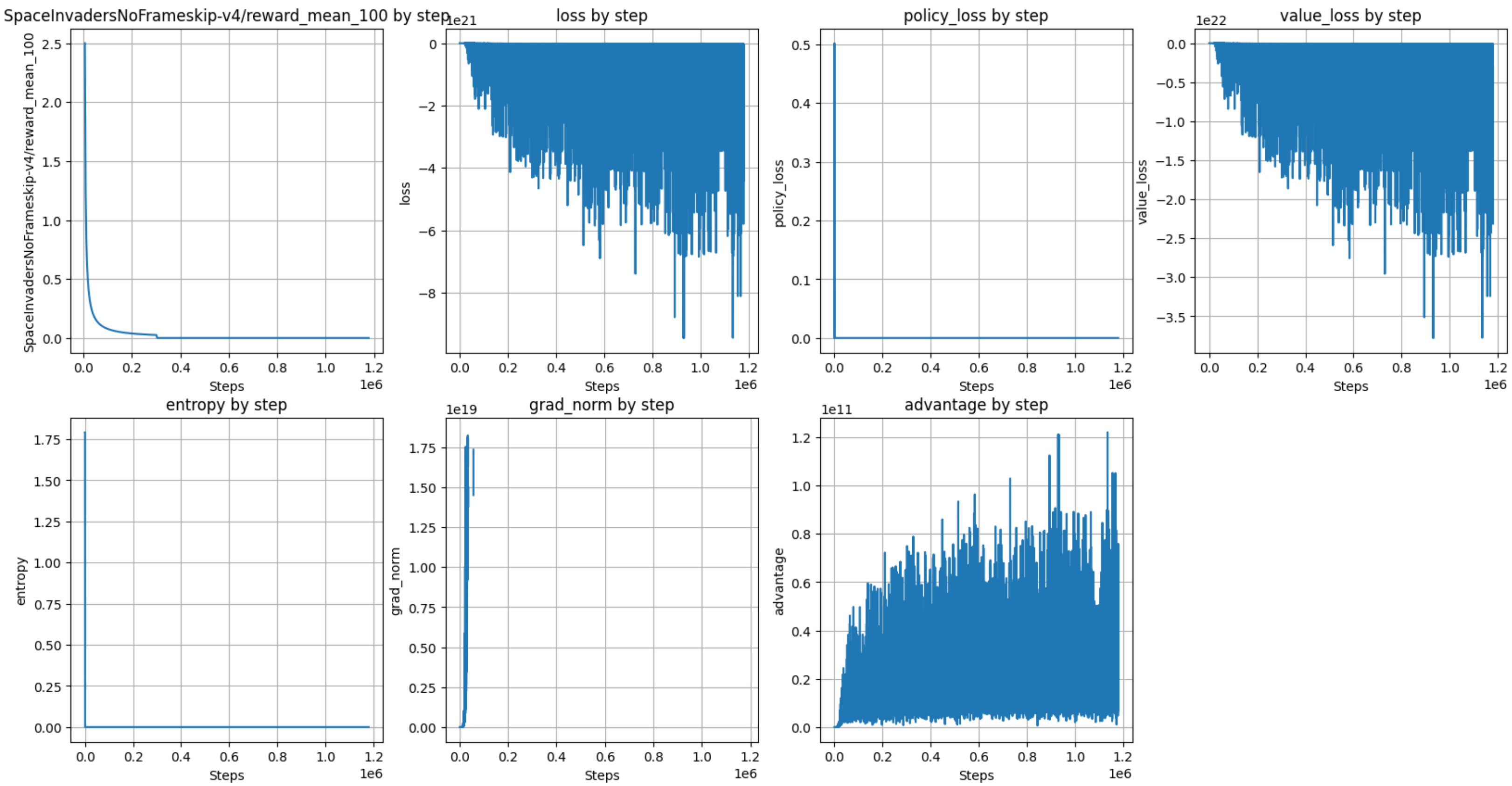
Example plots: correct solution 10M steps



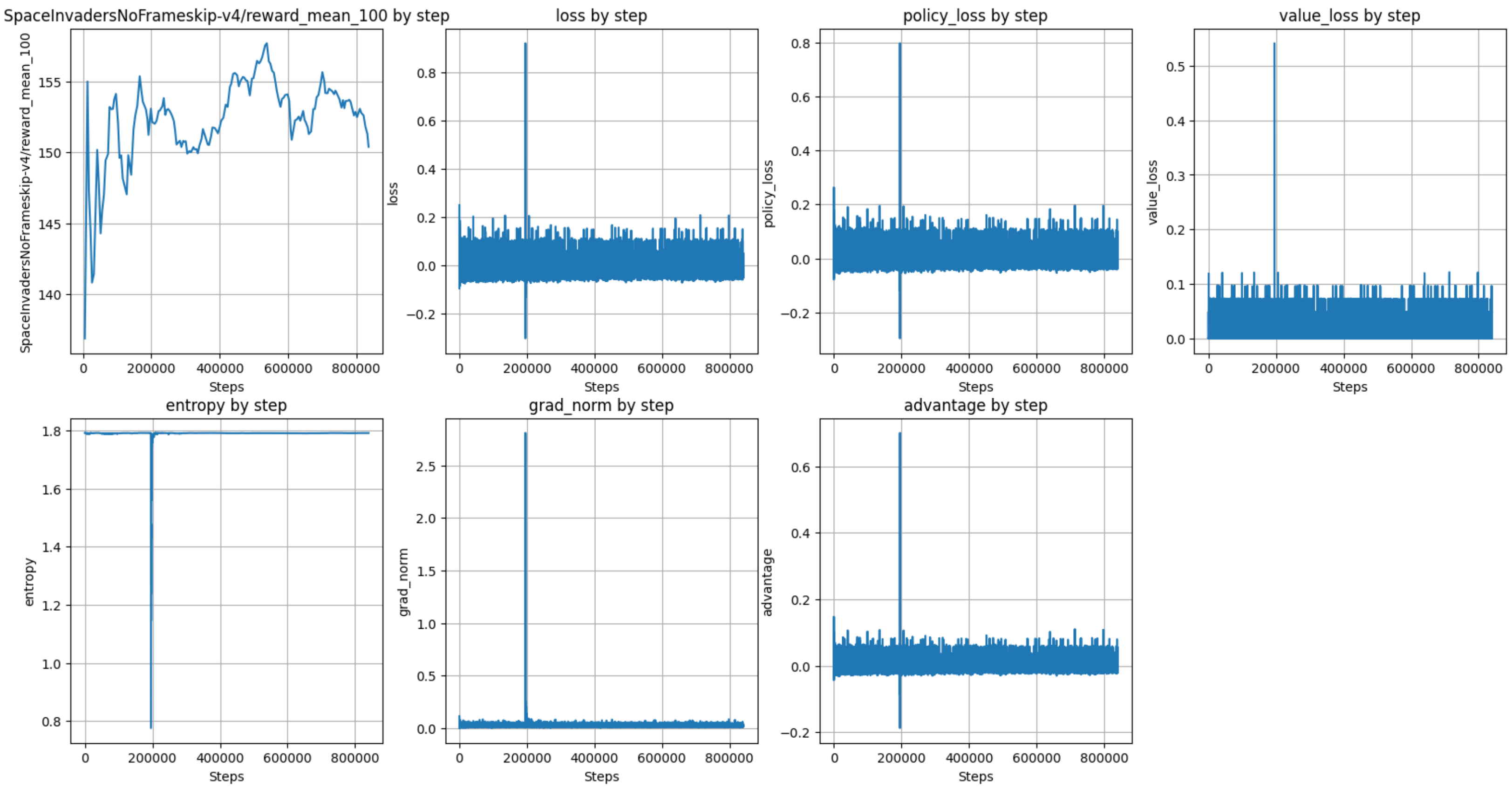
Example plots: bug in policy loss sign



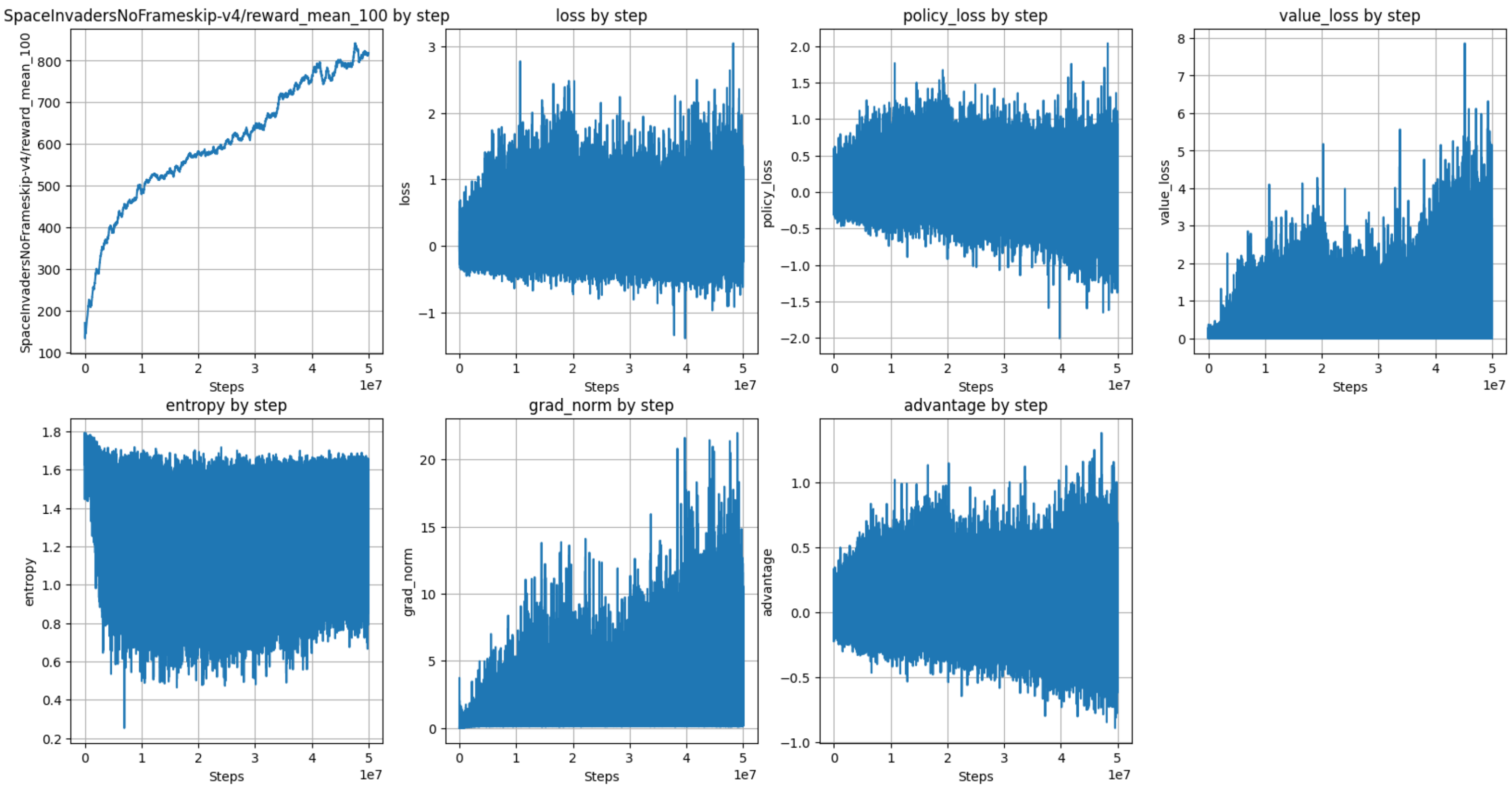
Example plots: bug in value loss sign



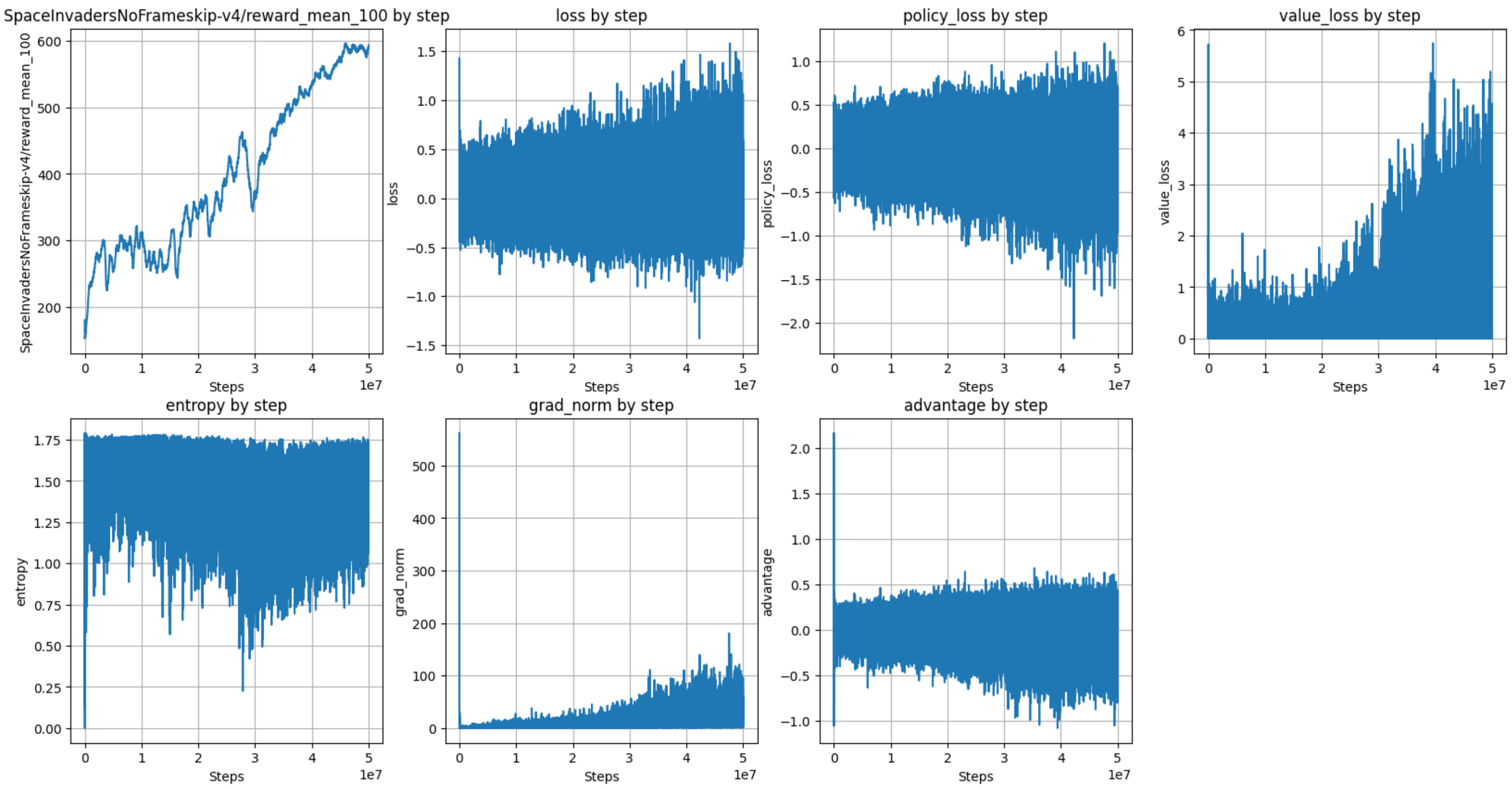
Example plots: inverted resets logic



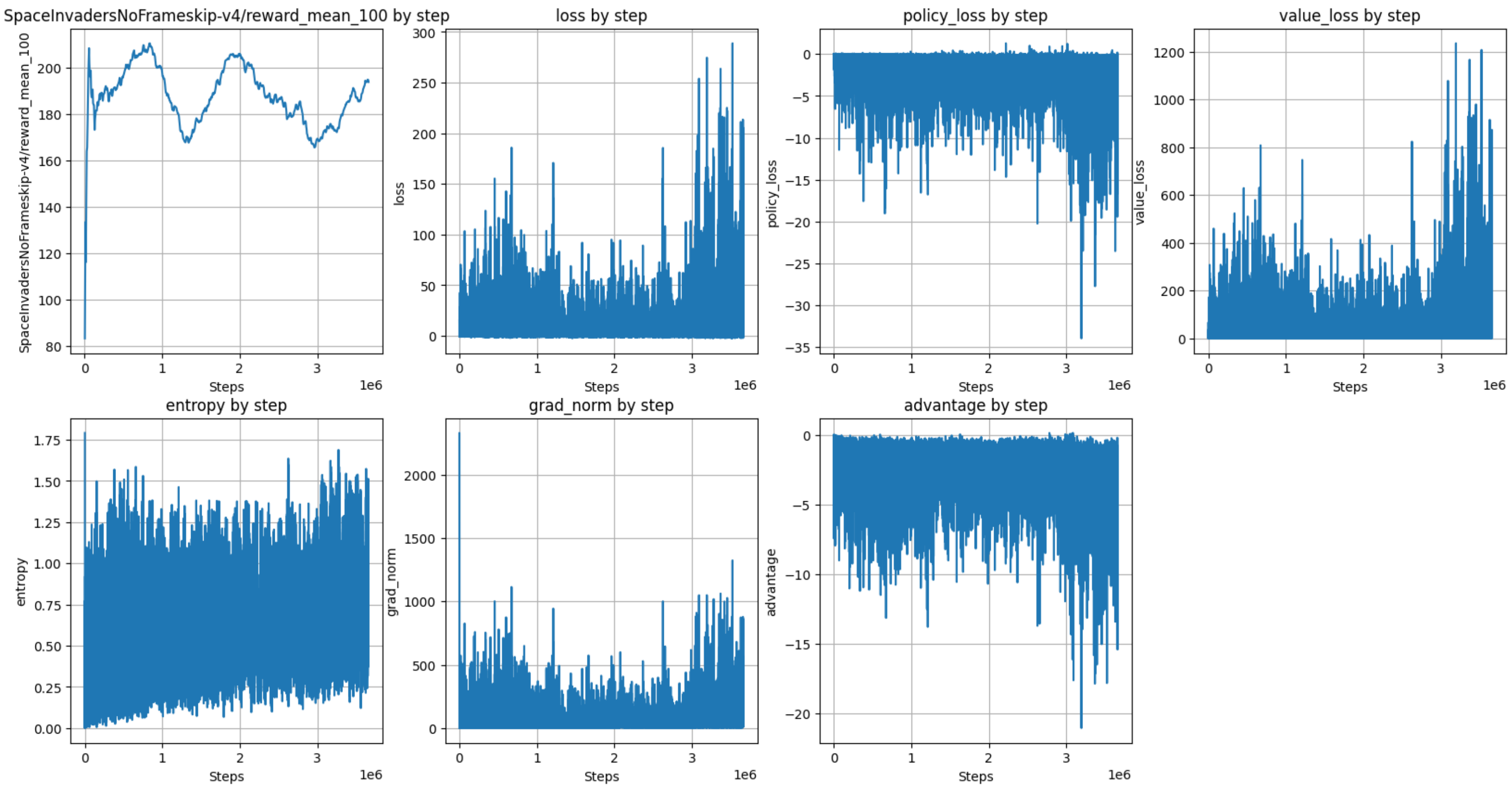
Example plots: no resets logic at all



Example plots: no img division by 255



Example plots: values grad in policy loss



Good luck!